



KCL

KVL

STAR-DELTA/DELTA-STAR
TRANSFORMATION

(Pradyumna K. Behera)

TYPES OF NETWORKS :-

- (1) Linear Circuit :- It is the circuit in which the parameters remain constant with change in applied voltage or current. Examples are the circuits containing resistors, inductors or capacitors. Ohm's law is applicable in this type of circuit.
- (2) Non-Linear Circuit :- It is a circuit whose parameters change with voltage or current. Examples are the circuits containing semiconductor devices like diodes. Ohm's law is not applicable here.
- (3) Unilateral Circuit :- It is a circuit, in which the properties or characteristics change with change in the direction of current flow. Examples are the circuits containing diodes, transistors, etc.
- (4) Bilateral Circuit :- It is a circuit whose properties remain same with the change in the direction of current flow. Examples are the circuits made of high conductive material elements.
- (5) Active Circuit :- It is a circuit which contains one or more voltage or current sources. Examples are the circuits containing batteries and transistors.
- (6) Passive Circuit :- It is a circuit which doesn't contain any source of energy. Examples are the circuits only containing passive elements like resistor, inductor or capacitor.
- (7) Lumped Circuit :- ~~Physically~~ The circuit containing physically separable elements like R, L & C is called lumped circuit.
- (8) Distributed Circuits :- The circuit in which, the circuit elements like R, L & C are not physically separated & exist in a distributed state is known as distributed circuit. Examples are the transmission line or cable circuits.
- (9) Recurrent Circuit :- When a large circuit consists of similar networks connected one after another, the circuit is called as Recurrent or cascaded circuit or ladder network.
- (10) Non-recurrent circuit :- A single network or circuit is called as a non-recurrent circuit.

* KIRCHHOFF'S LAWS :-

→ Kirchhoff's first law is ~~based~~ based on the law of conservation of charge. Charge can't be created but must be conserved.

(1) KIRCHHOFF'S CURRENT LAW (KCL) :-

→ It states that, "The sum of the currents at a node must equal zero." Math:- $\sum_{n=1}^N i_n = 0$ Where, N = No of branches connected to the node.

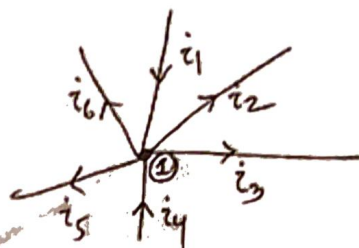
i_n = n th current entering (or leaving) the node.

→ By taking entering currents as positive & leaving currents as negative, apply KCL at node '1',

$$i_1 - i_2 - i_3 + i_4 - i_5 - i_6 = 0$$

$$\Rightarrow i_1 + i_4 = i_2 + i_3 + i_5 + i_6$$

→ The sum of the currents entering a node is equal to the sum of the currents leaving the node.



(Currents at a node showing KCL)

(2) KIRCHHOFF'S VOLTAGE LAW (KVL) :-

→ It is based on the principle of conservation of energy.

→ It states that, "the algebraic sum of all voltages around a closed path (or loop) is zero."

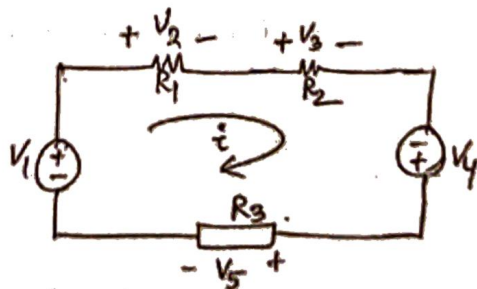
Math:- $\sum_{m=1}^M V_m = 0$ Where, M = No of voltages in the loop
 V_m = m th voltage.

→ By taking polarity convention & travelling in the clockwise direction, applying KVL to the closed loop:-

$$V_1 - iR_1 - iR_2 + V_4 - iR_3 = 0$$

$$\text{or } V_1 - V_2 - V_3 + V_4 - V_5 = 0$$

$$\Rightarrow V_1 + V_4 = V_2 + V_3 + V_5$$



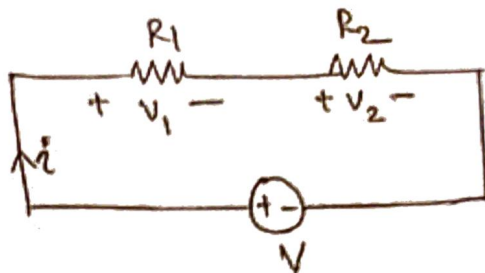
→ It can be also stated as, In a closed path, the sum of voltage rises is equal to the sum of voltage drops.

* VOLTAGE DIVISION RULE :-

→ In series connected resistors, voltage is divided & current remains same.

$$V = V_1 + V_2 = i(R_1 + R_2).$$

$$\Rightarrow i = \frac{V}{R_1 + R_2}.$$



Now, $V_1 = iR_1 = \left(\frac{R_1}{R_1 + R_2}\right) V.$

and $V_2 = iR_2 = \left(\frac{R_2}{R_1 + R_2}\right) V.$

→ In general, if there are 'N' resistors ($R_1, R_2, \dots, R_N$) are in series with a voltage source (V), the voltage across n th resistor (R_n) will have the value :-

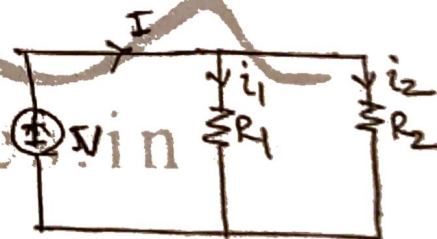
$$V_n = \frac{R_n}{R_1 + R_2 + \dots + R_N} \times V$$

* CURRENT DIVISION RULE :-

→ In parallel connected resistors, current is divided & voltage remains same.

$$I = i_1 + i_2 = \frac{V}{R_1} + \frac{V}{R_2} = V \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$\Rightarrow V = \frac{I}{\left(\frac{1}{R_1} + \frac{1}{R_2}\right)}$$



Now, $i_1 = \frac{V}{R_1} = \frac{I}{R_1 \left(\frac{1}{R_1} + \frac{1}{R_2} \right)} \Rightarrow i_1 = \frac{\left(\frac{1}{R_1}\right)}{\left(\frac{1}{R_1} + \frac{1}{R_2}\right)} I.$

Similarly, $i_2 = \frac{\left(\frac{1}{R_2}\right)}{\left(\frac{1}{R_1} + \frac{1}{R_2}\right)} \times I.$

→ In general, if there are 'N' resistors ($R_1, R_2, \dots, R_N$) are in parallel with source current (I), the current through n th resistor (R_n) is given by :-

$$i_n = \frac{\frac{1}{R_n}}{\left(\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}\right)} \times I$$

PROBLEMS :-

- ① find V_1 & V_2 in the circuit shown in the figure (1).

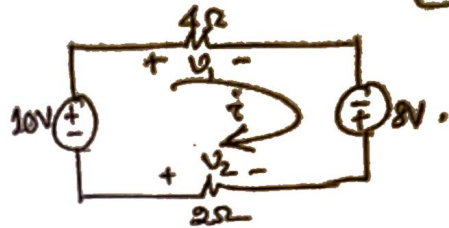
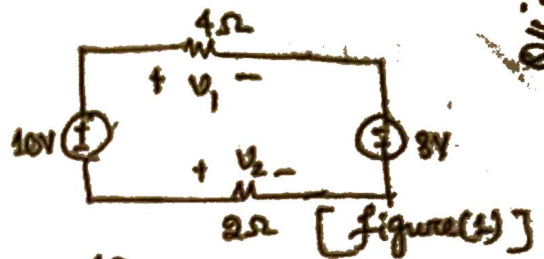
Solⁿ:- By applying KVL,

$$10 - 4i + 8 - 2i = 0$$

$$\Rightarrow 18 = 6i \Rightarrow i = 3A$$

$$V_1 = 4i = \boxed{12V}$$

$$V_2 = -2i = \boxed{-6V} \text{ (Ans)}$$



- ② Determine V_0 & i in the circuit shown in figure (2).

Solⁿ:- Applying KVL,

$$12 - 4i - 2V_0 + 4 - 6i = 0 \text{ --- (1)}$$

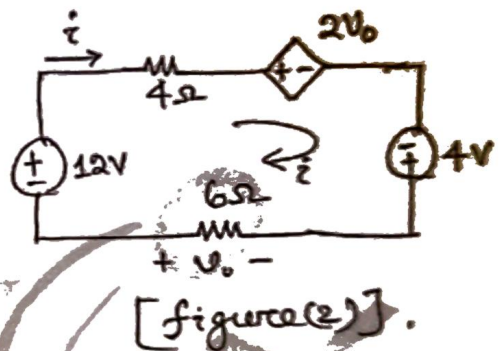
from Ohm's law, $V_0 = -6i$

Putting V_0 in eqⁿ (1) :-

$$12 - 4i + 12i + 4 - 6i = 0$$

$$\Rightarrow 2i = -16 \Rightarrow \boxed{i = -8A} \text{ (Ans)}$$

$$\therefore V_0 = -6 \times (-8) = \boxed{48V} \text{ (Ans)}$$

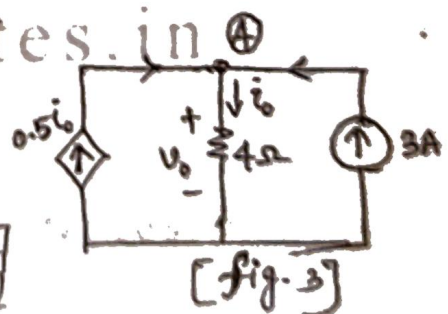


- ③ Find current i_0 & voltage V_0 in the circuit as shown in figure (3).

Solⁿ:- At node 'A', putting KCL :-

$$0.5i_0 + 3 = i_0 \Rightarrow 0.5i_0 = 3 \Rightarrow \boxed{i_0 = 6A} \text{ (Ans)}$$

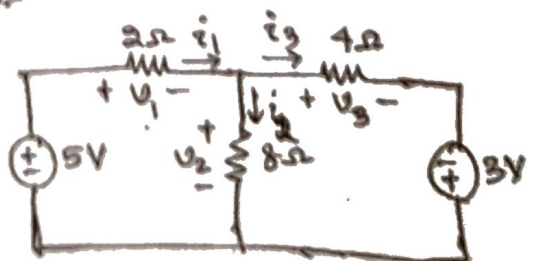
from Ohm's law, $V_0 = 4i_0 = \boxed{24V}$



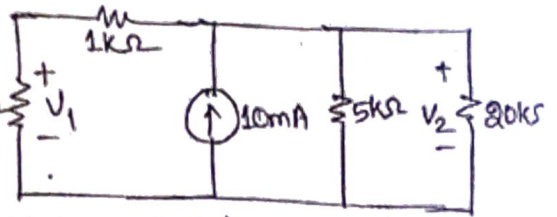
- ④ Find the currents & voltages in the circuit as shown in fig. (4).

Ans: $V_1 = 3V, V_2 = 2V, V_3 = 5V$

$i_1 = 1.5A, i_2 = 0.25A, i_3 = 1.25A$

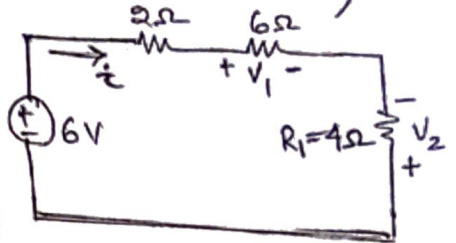


Q1: For the circuit shown in the figure, find (a) V_1 & V_2 (b) power dissipated in $3k\Omega$ & $20k\Omega$ resistors, and (c) power supplied by the current source.



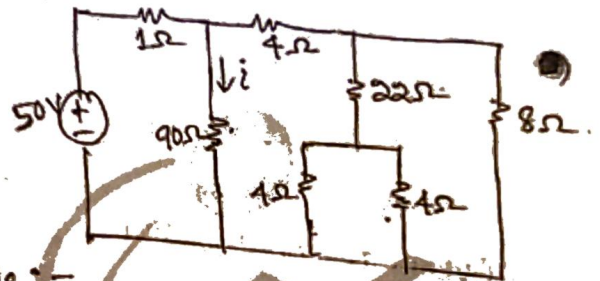
(Ans: (a) 15V, 20V; (b) 75 mW, 20 mW; (c) 200 mW.)

Q2: For the circuit shown in figure, find (a) The equivalent resistance seen by the source, (b) i , (c) power delivered by the source, (d) the voltages V_1, V_2 .



(Ans: (a) 10Ω , (b) 0.5A, (c) 3 watt, (d) $V_1 = 2V, V_2 = -2V$)

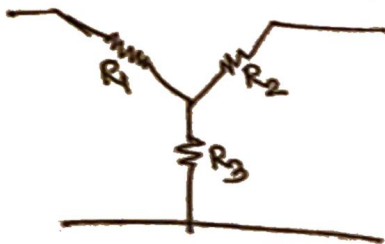
Q3: Find the equivalent resistance seen by the source and the current i in the circuit shown in figure.



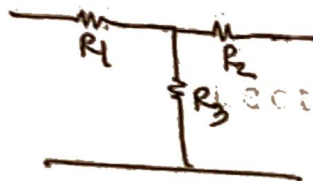
(Ans: $R_{eq} = 10\Omega, i = 0.5A$.)

* STAR(Y) - DELTA(Δ) TRANSFORMATIONS:

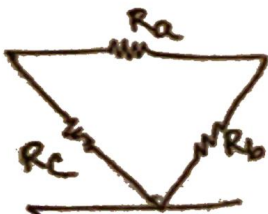
→ In the given circuit, it is very difficult to identify the resistors are connected in series or parallel. So, this type of circuit can be analysed or simplified into star(Y) and Delta(Δ) connection.



Star(Y)

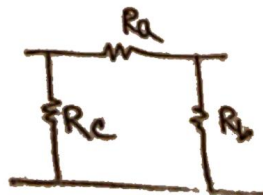


or Tee(T)

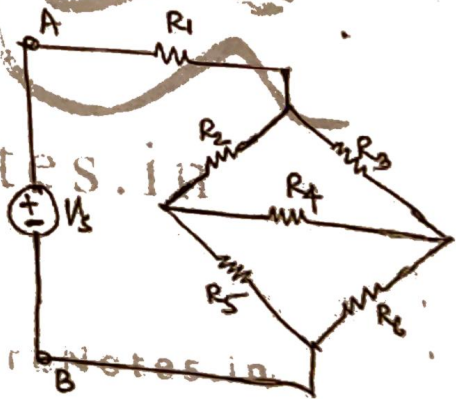


Delta(Δ)

or



Pi(Π)



(1) DELTA to STAR(Y) Conversion :- (Δ -Y) :-

Given :- R_a, R_b & R_c in Δ -Connection,
To find :- R_1, R_2 & R_3 to form Y-connection.

From figure,

$$R_{12}(Y) = R_1 + R_3$$

$$R_{12}(\Delta) = R_c \parallel (R_a + R_b)$$

By equating :-

$$R_{12}(Y) = R_{12}(\Delta)$$

$$\Rightarrow R_1 + R_3 = \frac{R_c(R_a + R_b)}{R_c + R_a + R_b} = R_{12} \quad (1)$$

Similarly :-

$$R_{13} = R_1 + R_2 = \frac{R_a(R_b + R_c)}{R_a + R_b + R_c} \quad (2)$$

$$R_{23} = R_2 + R_3 = \frac{R_b(R_c + R_a)}{R_b + R_c + R_a} \quad (3)$$

Subtracting eq(3) from eq(1) :-

$$R_1 - R_2 = \frac{R_c(R_a + R_b)}{R_c + R_a + R_b} - \frac{R_b(R_c + R_a)}{R_b + R_c + R_a} = \frac{R_a(R_c - R_b)}{R_a + R_b + R_c} \quad (4)$$

Adding eq(2) & eq(4) :-

$$R_1 = \frac{R_a R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a R_b}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_b R_c}{R_a + R_b + R_c}$$

→ Each resistor in the Y-network is the product of the resistors in the two adjacent Δ branches, divided by the sum of the three Δ resistors.

(2) STAR(Y) to Delta(Δ) Conversion :-

Given :- R_1, R_2 & R_3 in Y-network.

To find :- R_a, R_b & R_c to form Δ -network.

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

→ Each resistor in the Δ -network is the sum of all possible products of Y resistors taken two at a time, divided by the opposite Y resistors.

Note :- The Y & Δ networks are said to be balanced when $R_1 = R_2 = R_3 = R_Y$, $R_a = R_b = R_c = R_\Delta$. Under these conditions :-

$$R_Y = \frac{R_\Delta}{3} \quad \& \quad R_\Delta = 3 R_Y$$

[$\therefore R_Y$ is less than R_Δ].

Q Obtain the equivalent resistance R_{ab} for the circuit shown in figure & use it to find current i .

Solⁿ:- By taking the indicated points as it is in Y-connection. After converting into Δ -connection & redrawing it:-

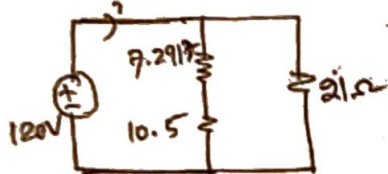
Let's take, $R_1 = 5\Omega$, $R_2 = 10\Omega$ & $R_3 = 20\Omega$.

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

$$\Rightarrow R_a = \frac{50 + 200 + 100}{20} = \frac{350}{20} = 17.5\Omega$$

$$\therefore R_b = \frac{350}{5} = 70\Omega, R_c = \frac{350}{10} = 35\Omega$$

By further reducing using series-parallel combinations:-



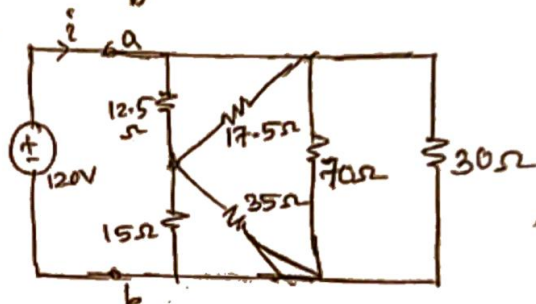
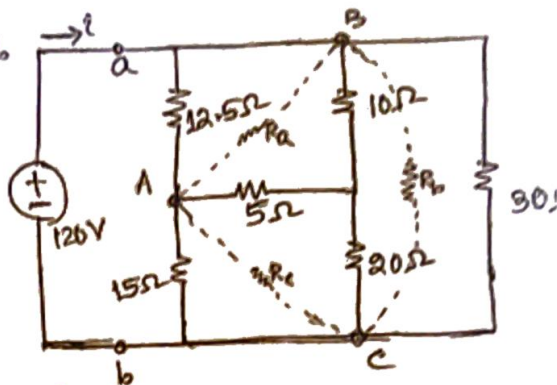
$$70 // 30 = \frac{70 \times 30}{100} = 21\Omega$$

$$12.5 // 17.5 = 7.2917\Omega$$

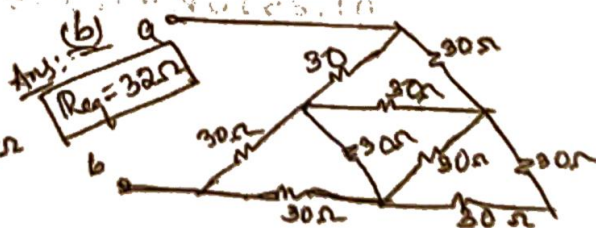
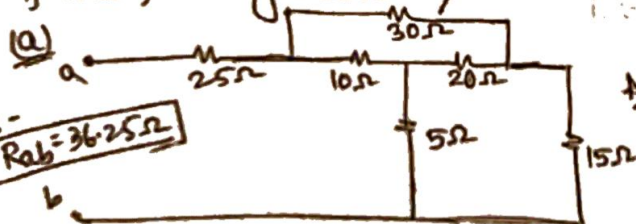
$$15 // 35 = 10.5$$

$$\therefore R_{ab} = (7.2917 + 10.5) // 21 = 9.632\Omega$$

$$\text{Now, } i = \frac{V}{R_{eq}} = \frac{V}{R_{ab}} = \frac{120}{9.632} = 12.458\text{ A. (Ans)}$$



Q Obtain the equivalent resistance at the terminals a-b for each of the following circuits.



Q Determine V in the circuit as shown in figure.

